

Suparno Saha

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TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C

Frontend: React, Next.js, React Native, TailwindCSS, shadcn/ui, React Flow, React Native Paper

Backend: Node.js, Express, Django, REST APIs, JWT Authentication, OAuth 2.0 (PKCE)

Databases: MongoDB, SQLite, Mongoose ODM

AI Integration: LangChain, Google Gemini API, Vector Embeddings, Prompt Engineering

Tools & Platforms: Git, Postman, Clerk Auth, Expo, Vercel, Render, PostHog

PROJECTS

NextWeeb | *React, TypeScript, TailwindCSS, Django, MyAnimeList API*

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- Engineered a **content-based recommendation system** that analyzes a user's full anime watch history (**1000+ entries**) and ranks new shows using multi-factor similarity scoring (genre compatibility, themes, score proximity, popularity tiers).
- Built a recommendation pipeline that fetches **500 popular anime candidates**, filters out watched titles, applies quality thresholds (score > 7.0), and detects sequels using regex to avoid redundant suggestions.
- Implemented secure authentication using **OAuth 2.0 PKCE** and migrated from session cookies to **stateless JWT auth**, resolving cross-domain login failures across production environments.
- Designed **15+ REST API endpoints** with protected routes, relational data models (User→AnimeEntry), list synchronization workflows, and aggregation logic powering personalized dashboards and viewing statistics.

Supernote | *Next.js, TypeScript, Express, MongoDB, LangChain, Clerk Auth*

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- Architected an **AI-driven ideation workflow** that transforms raw startup ideas into structured, multi-stage project plans covering market research, monetization, scalability, and technical complexity.
- Implemented **LangChain-based orchestration** with vector embeddings and structured prompt pipelines to generate context-aware node suggestions across interconnected idea graphs.
- Engineered an interactive **graph-based visualization system** with real-time node editing and dynamic AI expansions, enabling users to explore complex idea trees visually.
- Designed a modular backend with **MVC architecture**, Mongoose schemas for projects/ideas/documents, and middleware-based route protection powering collaborative workflows and Markdown-to-PDF synthesis.

Synapse | *React Native, Expo, Gemini API, AssemblyAI, YouTube API*

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- Engineered a **mobile study workflow** that converts lectures into structured learning materials, integrating speech-to-text transcription, OCR-based document scanning, and video transcript extraction into a unified content pipeline.
- Built an **AI-powered flashcard generation system** producing structured question-answer pairs from lecture transcripts and notes, designed for spaced repetition-based revision sessions.
- Designed hierarchical content management with groups and notebooks using persistent storage, enabling users to organize **100+ lecture notes** with editable transcripts and reusable study assets.
- Implemented cross-platform mobile architecture with modular navigation, custom UI states (loading, recording, editing), and real-time transcription flows for smooth multi-screen learning interactions.

ACHIEVEMENTS

- 1st Place — TechTrix Software Development Hackathon 2025:** Won overall by building Synapse, an AI learning app, and iterating features live based on judge feedback during finals.
- Top 3 — Blockchain Track, Hacksagon 2025:** Secured a podium finish in a specialized track at a national-level hackathon.
- Best Beginner Team — HackForge Hackathon:** Recognized for building a complete emergency response system consisting of **2 mobile apps and 1 web platform** within a 12-hour hackathon.
- 3rd Place — Creative Mode Track, IdeaTex Session Zero:** Designed and presented a Minecraft-based economic simulation modeling supply-demand through villager trading mechanics.

EDUCATION

Heritage Institute of Technology

Bachelor of Technology in Computer Science and Business System

2028

CGPA: 8.54